

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: ATOMIC AND LASER PHYSICS

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Wednesday, 07 October

Opening time: 09:30 am (BST)

You have two hours and thirty minutes to complete the paper and upload your answer file

*Answer **two** questions.*

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors is available at the Physics website.

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) Consider Schrödinger's equation for an N -electron atom:

$$\left[\sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i} \right) + \sum_{i>j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} \right] \psi = E\psi,$$

where m is the electron mass, Z is the atomic charge, $\mathbf{r}_i$ is position of the i -th electron and $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. Discuss what is meant by the *central field approximation* and discuss which are good quantum numbers that describe the electronic configuration. [5]

(b) By making reasonable assumptions at small and large distances from the atomic nucleus, show graphically how the potential may appear for the case of a sodium atom. [6]

(c) The electrons around the nucleus can be described *as if* they are a Fermi gas. In this approximation the number of electrons per unit volume with momentum less than p is $n = 8\pi p^3/3h^3$. Show that the combined potential due to all the electrons and the nucleus takes the form $V(r) = Z_{\text{eff}}e/4\pi\epsilon_0 r$ with the effective nuclear charge $Z_{\text{eff}} = \chi Z$ given by

$$\frac{d^2\chi}{dx^2} = \frac{\chi^{3/2}}{x^{1/2}},$$

where $x = r/b$. Determine an expression for b . [8]

(d) Discuss the limit of applicability of the above model. [2]

(e) Explain why in potassium the 4s level is filled before the 3d level. [4]

2. (a) Consider the approximation in which the central field and all the electrostatic and electromagnetic interactions internal to the electron system are included in the zeroth order Hamiltonian. The atom is in an external magnetic flux density $\mathbf{B}$ and the perturbation is written as

$$\mathcal{H} = A\mathbf{I} \cdot \mathbf{J} + g_J\mu_B\mathbf{J} \cdot \mathbf{B} - g_I\mu_N\mathbf{I} \cdot \mathbf{B}.$$

Describe the physical meaning of each term in this Hamiltonian. [3]

(b) Make a numerical estimate for the required field strength to be in the regime where the first term is dominant (the weak-field limit). [2]

(c) For the weak field case, show that the energy shifts induced by this Hamiltonian can be written as

$$\Delta E = \frac{1}{2}AK + g_F\mu_B BM_F,$$

and find an expression for K and g_F . [7]

(d) Draw the resultant energy level diagram for the case $J = 3/2$ and $I = 3$. [5]

(e) Now consider the perturbation Hamiltonian in the strong field case. Calculate the energy shifts in the ground state of hydrogen, $1s S_{1/2}$, and show that the following relation applies:

$$\frac{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=-1/2, I_z=-1/2}}{\Delta E_{J_z=1/2, I_z=-1/2} + \Delta E_{J_z=1/2, I_z=1/2}} = \frac{g_p\mu_N}{g_s\mu_B},$$

where g_p and g_s are the proton and electron Landé g -factors, respectively. [8]

3. (a) Discuss the propagation of light in a gain medium described by a two-level atomic system. Assume the population densities and degeneracies of the upper and lower levels are N_1 , g_1 and N_2 , g_2 , respectively. Also assume the levels are homogeneously broadened by a lineshape function $g(\omega - \omega_0)$, where ω_0 is the frequency separation of the two levels. The two levels are pumped at rates R_1 and R_2 ; and their corresponding lifetimes are τ_1 and τ_2 , respectively. Assume that the laser beam propagates in one dimension, along z . Take the spectral width of the laser intensity to be much smaller than the width of the transition line, that is, $I(z, \omega) = I(z)\delta(\omega - \omega_L)$, where ω_L is the laser line frequency. Show that in this particular system the laser intensity grows as a function of the distance, z , as

$$\frac{dI(z)}{dz} = \frac{\alpha_0 I(z)}{1 + I(z)/I_S},$$

and find an expression for α_0 and I_S in terms of the above quantities and the Einstein coefficients for the relevant transitions. [8]

(b) The result obtained in the previous section assumes one-dimensional propagation along z . However, this need not to be the case (as in many astrophysical lasers). A gain medium has the form of a sphere of radius R in which there is a uniform population inversion density. By considering the power gained as a spherical wave propagates outwards through a shell between r and $r + dr$, derive the appropriate equation that describes the growth of the intensity as a function of r . [4]

(c) The intensity coming out of the sphere is $0.1I_S$. Show that the power $P(r) = 4\pi r^2 I(r)$ grows exponentially within the sphere. Find an expression for $I(r)$. [4]

(d) Show that $I(r)$ has a minimum value and find the radius at which this occurs. [4]

(e) Comment on the value of $I(r)$ at $r = 0$. [5]

4. (a) Consider an atom with ground state energy E_1 and excited state energy E_2 interacting with a laser beam of frequency $\omega = \omega_0 + \delta\omega$, where $\omega_0 = (E_2 - E_1)/\hbar$, and $|\delta\omega/\omega_0| \ll 1$. Assume the laser beam is propagating in the z direction, and the amplitude of the laser electric field is $\mathcal{E}_0$. In the semi-classical picture, the laser interaction is described via a dipole potential $V(t) = ex\mathcal{E}_0 \cos(\omega t)$. Show that the solution of the time-dependent Schrödinger equation for this system can be written in the form

$$\begin{aligned}\dot{c}_1(t) &= i \frac{\mathcal{E}_0 \chi_{12}}{2\hbar} \left[e^{i(\omega - \omega_0)t} + e^{-i(\omega + \omega_0)t} \right] c_2(t), \\ \dot{c}_2(t) &= i \frac{\mathcal{E}_0 \chi_{12}}{2\hbar} \left[e^{-i(\omega - \omega_0)t} + e^{i(\omega + \omega_0)t} \right] c_1(t),\end{aligned}$$

where $|c_1(t)|^2$ and $|c_2(t)|^2$ are the probabilities of finding the atom in ground state or in the excited state, respectively, and

$$\chi_{ab} = -e \int \psi_a^* x \psi_b \, d\mathbf{r},$$

where $\psi_{a,b}$ ($a, b = 1, 2$) are the eigenstates of the unperturbed system. [6]

(b) Consider the case of a weak perturbation (that is, the atomic state does not change much). If the system is initially in the ground state, using the rotating-wave approximation, find an expression for $|c_2(t)|^2$. [6]

(c) Assume that the laser field is not exactly monochromatic, but it is specified by a spectral energy density $u(\omega)$, which satisfies the relation

$$\epsilon_0 \mathcal{E}_0^2 / 2 = \int_{\omega_0}^{\omega_0 + \Delta\omega} u(\omega) d\omega,$$

where $\Delta\omega$ is a small frequency interval. Show that the Einstein coefficient for absorption in a gas of atoms is given by

$$B_{12} = \frac{\pi \chi_{12}^2}{3\epsilon_0 \hbar^2}.$$

[4]

(d) Calculate the Einstein coefficient for the $1s \rightarrow 2p$ transition in hydrogen. The hydrogen wavefunctions can be written as $\psi_{n,l,m}(r, \theta, \varphi) = R_{n,l}(r)Y_{l,m}(\theta, \varphi)$, and they are tabulated below. [9]

n	l	m	$R_{n,l}$	$Y_{l,m}$
1	0	0	$2(1/a_0)^{3/2} \exp(-r/a_0)$	$(1/4\pi)^{1/2}$
2	0	0	$2(1/a_0)^{3/2} (1 - r/2a_0) \exp(-r/2a_0)$	$(1/4\pi)^{1/2}$
2	1	0	$(2/\sqrt{3})(1/a_0)^{3/2} (r/2a_0) \exp(-r/2a_0)$	$(3/4\pi)^{1/2} \cos \theta$
2	1	-1	$(2/\sqrt{3})(1/a_0)^{3/2} (r/2a_0) \exp(-r/2a_0)$	$(3/8\pi)^{1/2} \sin \theta e^{-i\varphi}$
2	1	+1	$(2/\sqrt{3})(1/a_0)^{3/2} (r/2a_0) \exp(-r/2a_0)$	$-(3/8\pi)^{1/2} \sin \theta e^{i\varphi}$